

Notes on Solid State Physics

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Foreword. This is a small project by Andias in an effort to learn condensed matter physics. "Teaching is the best way of learning" he thinks. This book is mainly based on Simon's "The Oxford Solid State Basics", but also takes inspiration from other books such as Ashcroft and Mermin's "Solid State Physics".

Prerequisites. I honestly don't know. I think readers should have some familiarity with classical mechanics, thermal/statistical physics, and quantum mechanics. This book was planned to be quite self-contained, meaning that every non-trivial prerequisites are going to be briefly explained in the first chapter (or along the way)

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1 Prerequisites

As I said in the foreword, I will have some pre-reqs given here:

1.1 Statistical Mechanics

The partition function is given by

1.2 Quantum Mechanics

2 Models of Matter

We will concern ourselves with the heat capacity of certain models of materials, comparing it to the Dulong-Petit Law and the experimental evidence of the cubic proportionality $C \sim T^3$ at low temperatures in many materials. We will also take a look at electrical conductivities.

2.1 Einstein model

Einstein treats matter as groups (or rather a “sea”) of free quantum harmonic oscillators that does not interact with one another. We will now formulate the heat capacity of matter in the Einstein model.

The energy of the quantum harmonic oscillator in one dimension, denoted as $E_n(\omega)$, is given by:

$$E_n(\omega) = \hbar\omega(n + 1/2)$$

One can then check that for a 3-dimensional harmonic oscillator, the energy $E_{n_1, n_2, n_3}(\omega)$ is given by:

$$E_{n_1, n_2, n_3}(\omega) = \hbar\omega(3/2 + n_1 + n_2 + n_3)$$

We can then construct a partition function for this system:

$$\begin{aligned} Z &= \sum_{n_1, n_2, n_3} e^{-\beta E_{n_1, n_2, n_3}} \\ &= \sum_{n_1, n_2, n_3} e^{-\beta \hbar\omega(3/2 + n_1 + n_2 + n_3)} \\ &= e^{-3\beta \hbar\omega/2} \sum_{n_1, n_2, n_3} e^{-\beta \hbar\omega(n_1 + n_2 + n_3)} \\ &= (2 \sinh(\beta \hbar\omega/2))^{-3} \end{aligned}$$

From this, we can find that the expected energy of the system to be

$$\begin{aligned} \langle E \rangle &= -\frac{1}{Z} \frac{\partial Z}{\partial \beta} \\ &= \frac{3}{2} \hbar\omega \coth(\beta \hbar\omega/2) \end{aligned}$$

Actually, we can rewrite the expression above as

$$\langle E \rangle = 3\hbar\omega(n_B(\beta \hbar\omega) + 1/2)$$

Where n_B is the Bose occupation factor:

$$n_B(x) = \frac{1}{e^x - 1}$$

The Bose occupation factor can be thought of as “how many particles are in a certain state.”

Why the Bose factor? Basically, we're treating quantum harmonic oscillators as bosons, or integer-spin particles. Thus we're also assuming that Pauli exclusion principle doesn't apply to these oscillators (which it definitely does for electrons, as we would see in the later parts of this chapter.) The concept of occupation factors will become important later on (mainly to simplify equations) when we're considering different types of particles, such as non-bosons (fermions).

Now, we will see the expression for heat capacity. Heat capacity is defined as the amount of change in energy for a certain amount of change in temperature. Thus, also remembering that $\beta = k_B T$, we find that:

$$C = \frac{\partial \langle E \rangle}{\partial T} = 3k_B(\beta\hbar\omega)^2 \frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}$$

which is quite a gruesome expression.

Now, let's see what happens in two cases: the high temperature limit and the low temperature limit.

- In the high temperature limit, we can consider $k_B T \gg \hbar\omega$. Then, $\beta\hbar\omega \rightarrow 0$. Expanding the Taylor series for the exponential terms in C gives us

$$C \approx 3k_B(\beta\hbar\omega)^2 \frac{1}{(\beta\hbar\omega)^2} = 3k_B$$

which perfectly fits the Dulong-Petit Law. As such, the Einstein model is accurate in predicting heat capacities in materials with low oscillation frequency.

- In the low temperature limit, however, the heat capacity quickly drops to zero due to the exponential term in the denominator. It doesn't follow the cubic correspondence seen in experiments.

The Einstein model has some flaws, mainly because it only considers one common frequency for every particles. We only consider one value of ω , which causes the predicted "die-off" in low temperatures according to the model. Fortunately there is a better way to do things, in the form of the Debye model.

2.2 Debye model

In the Debye model, we're not assuming that all oscillators oscillate at the same frequencies, but rather we're considering a wide range of normal modes. Essentially, Debye considers that atomic oscillations should be quantized the same way as light is quantized, but with the difference being that light has two polarizations that corresponds to the transversal modes, whereas the atomic oscillations have three, with the extra being the longitudinal mode. For the sake of simplicity, however, we consider the three polarizations to be propagating with the same velocity (which is not really true most of the time).

For the derivations, we implement Born-von-Karman boundary conditions. Simply put, we consider that the atomic oscillations are waves that are periodic-L. In one dimension, we can think of it as a function lying on a circular string with circumference L . The position repeats every length L , and for a wavefunction ψ on the string at position r ,

$$\psi(r) = \psi(r + L)$$

is assumed to hold true.

If we consider the usual form for waves which is $\psi(r) = e^{ikr}$ where k is the wave vector, we get that the Born-von-Karman condition imply

$$e^{ikL} = 1 \implies kL = 2\pi n \quad \text{where } n \in \mathbb{N}$$

or in a nicer form,

$$k = \frac{2\pi n}{L}$$

In 3-dimensions, we basically have that in a cube with vertices L ,

$$\vec{k} = \frac{2\pi}{L}(n_1, n_2, n_3)$$

where n_1, n_2, n_3 corresponds to the 3 different directions that are perpendicular to each other (which is basically x, y, and z if you're a bit lazy to think).

Debye's idea was that the frequency of oscillation is a function of the wave vector, and that we should sum over all possible $\vec{k}$ to get the right total energy of the system. Einstein has given us the energy expression for one oscillation frequency, and thus to get the corrected value,

$$\langle E \rangle = \sum_{\vec{k}} 3\hbar\omega(n_B(\beta\hbar\omega) + 1/2)$$

Evaluating sums like this might not be a good idea to do directly, so I will introduce a small trick to simplify our derivation.

Let us go back to the one-dimensional case. In k -space, basically each points of k occupies a space of “length” $2\pi/L$ on the line. As L becomes large, we have that each k -points behave like points in $\mathbb{R}$. Hence, we can transform the sum into

$$\sum_k \rightarrow \frac{L}{2\pi} \int_{-\infty}^{+\infty} dk$$

which is a better thing to work with than sums.

The same idea can be applied to the three-dimensional case. Instead of occupying a line-space, each values of $\vec{k}$ occupies a volume-space, which can be described as a cube having vertices of length $2\pi/L$. As such, the cube has volume $(2\pi/L)^3$. Using the same analogy, we get that the sum becomes

$$\sum_{\vec{k}} \rightarrow \frac{L^3}{(2\pi)^3} \int d^3k$$

where I have used the notation $\int d^3k = \iiint dk_1 dk_2 dk_3$ for simplicity. Basically, a volume integral over the three dimensions in k -space.

Invoking spherical symmetry in k -space, the integral can be transformed into

$$4\pi \int_0^\infty dk \ k^2$$

where this k in the integral is a magnitude of the vector $\vec{k}$ that we had earlier (the notation can be a bit confusing at times, I know, I feel you). Intuition-wise, this is like a ball with radius $|\vec{k}|$.

Now, our expression for energy becomes

$$\langle E \rangle = \left(\frac{L}{2\pi}\right)^3 \int d^3k \ 3\hbar\omega(n_B(\beta\hbar\omega) + 1/2) = 3 \frac{4\pi L^3}{(2\pi)^3} \int_0^\infty dk \ k^2 \hbar\omega(n_B(\beta\hbar\omega) + 1/2)$$

which is also quite a gruesome expression.

A key feature of the Debye model is it's assumed that the oscillation frequency follows the dispersion relation $\omega(\vec{k}) = v|\vec{k}| = vk$ where v is the speed of propagation of the wave. We can then substitute $k = \omega/v$ to our expression and get

$$\langle E \rangle = 3 \frac{4\pi L^3}{(2\pi)^3} \int_0^\infty \omega^2 d\omega (1/v^3) \hbar\omega(n_B(\beta\hbar\omega) + 1/2)$$

We now introduce a concept called “density of states”, which we will denote by $g(\omega)$. Basically, $g(\omega)d\omega$ is the amount of allowed states between ω and $\omega + d\omega$. In this case, we define the density of states as

$$g(\omega) = 3 \frac{4\pi L^3}{(2\pi)^3} \frac{\omega^2}{v^3}$$

We can do a bit of more simplification. Using the fact that in a box of volume L^3 that contains N particles, we define the density $n = N/L^3$. Also, we define something called the Debye frequency, which is $\omega_D = (6\pi^2 n v^3)^{1/3}$. We now get

$$g(\omega) = N \frac{9\omega^2}{\omega_D^3}$$

After this, we can substitute it to the expression for energy. We will then get that

$$\langle E \rangle = \int_0^\infty g(\omega) d\omega \quad \hbar\omega (n_B(\beta\hbar\omega) + 1/2)$$

An intuitive view for processing all of this is that basically the integral says: “The total energy of the system is the sum over energy-per-state (all of the terms on the right of $d\omega$ in the integral) all possible states (implied by the density of states function)”. I think this is quite cool, and a nice way to view complicated expressions in physics.

Now, let’s actually try and evaluate the integral. Subbing the density of states back to the integral gives us

$$\langle E \rangle = \frac{9N\hbar}{\omega_D^3} \int_0^\infty d\omega \quad \omega^3 (n_B(\beta\hbar\omega) + 1/2) = \frac{9N\hbar}{\omega_D^3} \left[\int_0^\infty d\omega \quad \omega^3 n_B(\beta\hbar\omega) + \int_0^\infty d\omega \quad \frac{\omega^3}{2} \right]$$

Notice that the second integral will evaluate to infinity, which is definitely not a great sign. I will denote the infinity as E_{inf} , but the term E_{zero} might be more useful for later concepts. This term is not dependent of temperature.

Let’s try and compute the heat capacity. First we’ll evaluate the energy:

$$\begin{aligned} \langle E \rangle &= \frac{9N\hbar}{\omega_D^3} \left[\int_0^\infty d\omega \quad \omega^3 n_B(\beta\hbar\omega) + \int_0^\infty d\omega \quad \frac{\omega^3}{2} \right] \\ &= \frac{9N\hbar}{\omega_D^3} \int_0^\infty \frac{\omega^3}{e^{\beta\hbar\omega} - 1} d\omega + E_{inf} \\ &= \frac{9N\hbar}{\omega_D^3 (\beta\hbar)^4} \int_0^\infty \frac{(\beta\hbar\omega)^3}{e^{\beta\hbar\omega} - 1} d(\beta\hbar\omega) + E_{inf} \\ &= \frac{9Nk_B^4 \pi^4}{15(\hbar\omega_D)^3} T^4 + E_{inf} \end{aligned}$$

Notice that if we differentiate it one time with respect to temperature, we get that $C \sim T^3$. The Debye model is accurate for low temperatures but at high temperatures, the heat capacity blows up. It doesn't fit the Dulong-Petit law, so we'll need a little fix to the model.

We will introduce a cut-off frequency, which we'll denote by ω_c . Debye postulated that the amount of normal modes should be the same as the degree of freedom in our object. In this case, we have N particles with 3 motion direction. Hence, we should have a total normal mode of $3N$. Mathematically, it's described with the integral of density of state:

$$3N = \int_0^{\omega_c} g(\omega) d\omega$$

Let's check the high-temperature limit. Here, $k_B T \gg \hbar \omega$, therefore $n_B(\beta \hbar \omega) \rightarrow (\beta \hbar \omega)^{-1}$. Substituting this back into the equation gives us

$$\begin{aligned} \langle E \rangle &= \int_0^{\omega_c} g(\omega) d\omega \hbar \omega \left(\frac{k_B T}{\hbar \omega} + 1/2 \right) \\ &= \int_0^{\omega_c} g(\omega) d\omega k_B T + \frac{1}{2} \int_0^{\omega_c} g(\omega) d\omega \hbar \omega \\ &= 3N k_B T + E_{zero} \end{aligned}$$

Notice that this "tweak" we imposed gives us the Dulong-Petit law. Also notice that by imposing a cut-off, we got rid of the infinity (hence why I wrote it as E_{zero} now instead of E_{inf}).

Now one may think, what is the value of the cut-off frequency? We can try and evaluate it from our initial assumption, which is that the total normal mode is the same as the degree of freedom. Let's evaluate the integral:

$$3N = \int_0^{\omega_c} g(\omega) d\omega = \int_0^{\omega_c} \frac{9N\omega^2}{\omega_D^3} d\omega = 3N \frac{\omega_c^3}{\omega_D^3} \implies \omega_c = \omega_D$$

Wow, it turns out that the cut-off frequency is our proposed Debye frequency!

2.3 Drude model

The Drude model of conductivity treats metals as a sea of electrons that scatter off of each other. This is a direct application of kinetic theory. Before doing any calculations, we must impose a few assumptions:

- The average time between collisions, or called *scattering time*, is denoted by τ . The probability of an electron being scattered in a time interval dt is given by $\frac{dt}{\tau}$
- After scattering happens, the electron becomes stationary
- In between scattering events, electrons respond to external forces (which in the cases we will see, the $\vec{E}$ and $\vec{B}$ fields)

With these assumptions in play, we can now formulate the Drude model.

We have an electron (charge $-e$) that has momentum $\vec{p}(t)$ at time t moving in the presence of a force $\vec{F}$. I will denote the probability of a particular event S as π_S . After a time interval dt , the average momentum of the electron becomes:

$$\langle \vec{p}(t + dt) \rangle = \sum_S \pi_S \vec{p}_S(t + dt) = \frac{dt}{\tau} \vec{0} + (1 - \frac{dt}{\tau})(\vec{p}(t) + \vec{F}dt)$$

The first term corresponds to the scattering event, and the second term corresponds to when scattering doesn't happen. When scattering doesn't happen, as we have stated in our initial assumption, the electron is affected by external forces. We now have

$$\langle \vec{p}(t + dt) \rangle = \vec{p}(t) - \frac{dt}{\tau} \vec{p}(t) + \vec{F}dt + O(dt^2)$$

As dt is taken to be infinitesimally small, we can ignore the quadratic term in the end. Subtracting both sides by $\vec{p}(t)$ and dividing the equation by dt , we will get that

$$\frac{\langle \vec{p}(t + dt) \rangle - \vec{p}(t)}{dt} = \frac{d\vec{p}}{dt} = \vec{F} - \frac{\vec{p}(t)}{\tau}$$

This is the jist of the whole model. Usually, we would want to look how the system behaves under the influence of electric and magnetic fields. Thus, we can say that the force is given by the Lorentz force $\vec{F} = -e(\vec{E} + \vec{v} \times \vec{B})$. We will see some special cases.

First, the case where magnetic field is absent. This might interest us as this is (approximately) what happens in wires. In the absence of magnetic field, the equation becomes

$$\frac{d\vec{p}}{dt} = -e\vec{E} - \frac{\vec{p}}{\tau}$$

It may be useful to take a look at the steady state of the function, or in other words, $\frac{d\vec{p}}{dt} = 0$. Steady state conditions gives us

$$\vec{p} + e\tau\vec{E} = 0 \implies \vec{v} + \frac{e\tau}{m}\vec{E} = 0 \implies \vec{v} = -\frac{e\tau}{m}\vec{E}$$

Recall from electrodynamics, the current density is given by $\vec{J} = -ne\vec{v}$ where n is the number of electrons per volume. Inputting the expression we got from the force calculation to the current density gives us

$$\vec{J} = \frac{ne^2\tau}{m}\vec{E}$$

This implies Ohm's Law where the conductivity $\sigma = \frac{ne^2\tau}{m}$, cool!

Now, what happens when there is a magnetic field? Let's again, but more carefully, take a look at the steady state condition. The momentum equations now give us

$$0 = e\vec{E} + e\vec{v} \times \vec{B} + \frac{\vec{p}}{\tau} = \frac{e\tau}{m}\vec{E} + \frac{e\tau}{m}(\vec{v} \times \vec{B}) + \vec{v}$$

To make things simpler, we can represent the velocity, electric field, and magnetic field as column vectors. More explicitly,

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}, \quad \vec{E} = \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix}, \quad \vec{B} = \begin{pmatrix} B_1 \\ B_2 \\ B_3 \end{pmatrix}$$

One can also check that the cross product of velocity and magnetic field can be represented as the matrix multiplication of a 3×3 magnetic field matrix and the column vector of velocity:

$$\vec{v} \times \vec{B} = \begin{pmatrix} v_2B_3 - v_3B_2 \\ v_3B_1 - v_1B_3 \\ v_1B_2 - v_2B_1 \end{pmatrix} = \begin{pmatrix} 0 & B_3 & -B_2 \\ -B_3 & 0 & B_1 \\ B_2 & -B_1 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

Hence, by introducing $\lambda = \frac{e\tau}{m}$, we can write the momentum equation as

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \lambda \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix} + \begin{pmatrix} 1 & \lambda B_3 & -\lambda B_2 \\ -\lambda B_3 & 1 & \lambda B_1 \\ \lambda B_2 & -\lambda B_1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

Now, by making the form a bit nicer we can get that

$$\vec{E} = \mathbf{B}\vec{v}$$

where $\mathbf{B}$ is what I would call the *scaled magnetic field matrix* given by

$$\mathbf{B} = \begin{pmatrix} -1/\lambda & -B_3 & B_2 \\ B_3 & -1/\lambda & -B_1 \\ -B_2 & B_1 & -1/\lambda \end{pmatrix}$$

This form implies that the conductivity is now a rank-2 tensor. Conductivity is not necessarily isotropic as in the case where magnetic field is absent.

Actually, we can see something interesting. Let us impose the conditions that the electric field only exists in the x-direction, and the magnetic field only exists in the y direction. This will give us

$$\begin{pmatrix} E \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/\lambda & 0 & B \\ 0 & -1/\lambda & 0 \\ -B & 0 & -1/\lambda \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} v_3 B - v_1/\lambda \\ -v_2 \lambda \\ -v_1 B - v_3/\lambda \end{pmatrix}$$

Solving the three equations will give us the expression for v_3

$$v_3 = \frac{E}{B} \left(1 - \frac{1}{\lambda^2 B^2} \right)$$

What is the significance of this? Notice that in this condition, there are 2 flows of charge: one parallel to the electric field, and one perpendicular to the electric field. This is precisely the Hall effect! Although, v_3 can be zero if $\lambda = B^{-1}$.